Problem 1. *Provide a parameterization $\vec{r}(t)$ of the curve C formed by the intersection of the circular cylinder of radius 1 about the z-axis, and the plane $z = y$. The parametrization must start at $\vec{r}(0) = (1, 0, 0)$ and end at the point $(0, 1, 1)$.*

Solution. The parameterization is

$$\vec{r}(t) = [\cos(t), \sin(t), \sin(t)].$$

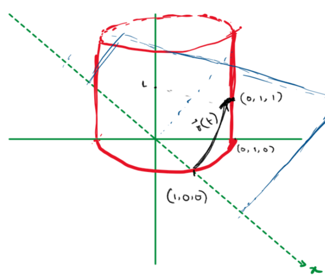

Figure 1: Sketch of the Curve.

□

Problem 2. *Sketch vector arrows to depict the vector field*

$$\vec{v} = -y\hat{\boldsymbol{i}} + x\hat{\boldsymbol{j}}.$$

Solution. Here is a depiction of the vector field.

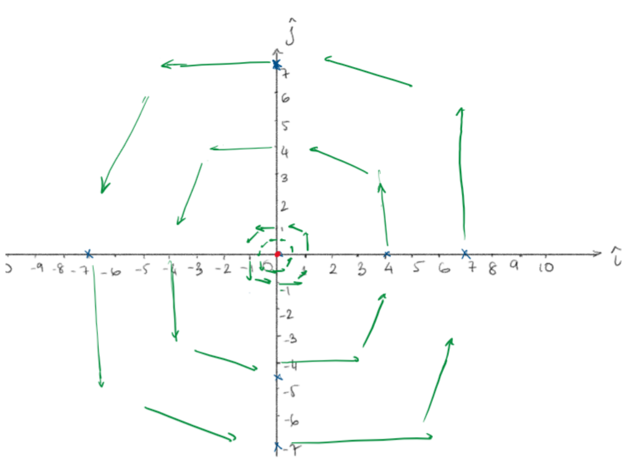

Figure 2: Vector Field.

Note: The size of arrows must be increasing as we go further away from the origin, and the directions should result in a counterclockwise circulation. □

Problem 3. *Evaluate* $\iiint_T (x+y+z)\, dxdydz$ *where* T *is the volume of the first octant inside the cone* $x^2+y^2 \leq 4z^2$, $z \in [0,2]$, *as shown below.*

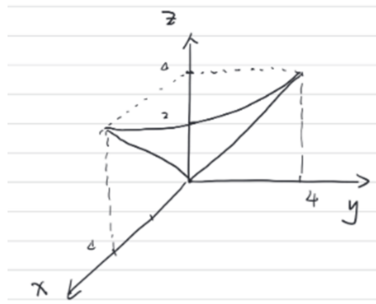

Figure 3: Region T.

Solution. Using cylindrical coordinates, the cone is $r^2 \leq 4z^2$. As we are in the first octant, $0 \leq \theta \leq \frac{\pi}{2}$. As $0 \leq z \leq 2$, then the largest possible radius would be $r^2 = 4(2)^2 = 16$, so $0 \leq r \leq 4$. In other words, the region T is described by:

$$0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq r \leq 2z, \quad 0 \leq z \leq 2.$$

Now, in cylindrical coordinates: $x = r\cos(\theta)$, $y = r\sin(\theta)$, and the Jacobian is r, so we get:

$$\begin{aligned}
\iiint_T (x+y+z)\, dxdydz &= \int_0^{\pi/2}\int_0^2\int_0^{2z} (r\cos(\theta)+r\sin(\theta)+z)\, r drdzd\theta \\
&= \int_0^{\pi/2}\int_0^2\int_0^{2z} (r^2\cos(\theta)+r^2\sin(\theta)+zr)\, drdzd\theta \\
&= \int_0^{\pi/2}\int_0^2 \left[\frac{1}{3}r^3(\cos(\theta)+\sin(\theta))+\frac{1}{2}zr^2\right]_0^{2z} dzd\theta \\
&= \int_0^{\pi/2}\int_0^2 \frac{1}{3}(2z)^3(\cos(\theta)+\sin(\theta))+\frac{1}{2}z(2z)^2\, dzd\theta \\
&= \int_0^{\pi/2}\int_0^2 \left[\frac{8}{3}z^3(\cos(\theta)+\sin(\theta))+2z^3\right] dzd\theta \\
&= \int_0^{\pi/2} \left[\frac{8}{3}\frac{z^4}{4}(\cos(\theta)+\sin(\theta))+2\frac{z^4}{4}\right]_0^2 d\theta \\
&= \int_0^{\pi/2} \left[\frac{2}{3}(2^4)(\cos(\theta)+\sin(\theta))+\frac{1}{2}(2)^4\right]; d\theta \\
&= \left[\frac{2^5}{3}(\sin(\theta)-\cos(\theta))+(2^3)\theta\right]_0^{\pi/2} \\
&= \frac{2^5}{3}((\sin(\pi/2)-\cos(\pi/2))-(\sin(0)-\cos(0)))+8\frac{pi}{2} \\
&= \frac{32}{3}((1-0)-(0-1))+4\pi \\
&= \frac{64}{3}+4\pi.
\end{aligned}$$

$\square$

Problem 4. *Verify Stokes' Theorem for*

$$\vec{F} = [y^3, -x^3, 0],$$

and S is the surface

$$x^2 + y^2 \leq 1, \quad z = 0.$$

Hint: To save time, you may directly use, if needed, the following facts:

1. *S can be parameterized as $x = u\cos v$, $y = u\sin v$, for $0 \leq u \leq 1$, $0 \leq v \leq 2\pi$, and in this set up, $\vec{N} = \vec{r}_u \times \vec{r}_v = u\hat{k}$.*
2. *The boundary ∂S, can be parameterized as $x(t) = \cos(t)$, $y(t) = \sin(t)$, $0 \leq t \leq 2\pi$.*
3. *The expression $\sin^4(t) + \cos^4(t)$ can be simplified to $\frac{3}{4} + \frac{1}{4}\cos(4t)$.*

Solution. Since the $\hat{k}$-component of $\vec{F}$ is zero, then

$$\nabla \times \vec{F} = \left(\frac{\partial(-x)^3}{\partial x} - \frac{\partial(y)^3}{\partial y} \right) \hat{k} = -3(x^2 + y^2)\hat{k}.$$

Then, on the surface with the given parameterization:

$$\nabla \times \vec{F}|_S = -3(u^2)\hat{k}.$$

Consequently,

$$\begin{aligned} iint_S(\nabla \times \vec{F}) \cdot \hat{n}\, dA &= \int_0^{2\pi} \int_0^1 (\nabla \times \vec{F}) \cdot \vec{N}\, dudv \\ &= \int_0^{2\pi} \int_0^1 (-3u^2\hat{k}) \cdot (u\hat{k})\, dudv \\ &= \int_0^{2\pi} dv \int_0^1 -3u^3\, du \\ &= (2\pi) \left[-3\frac{u^4}{4} \right]_0^1 \\ &= -\frac{6\pi}{4} = -\frac{3}{2}\pi. \end{aligned}$$

On the other hand, using the given parameterization $\vec{r}(t) = [\cos(t), \sin(t), 0]$, we get

$\vec{r}'(t) = [-\sin(t), \cos(t), 0]$ and on the boundary, $\vec{F}|_{\partial S} = [\sin^3(t), -\cos^3(t), 0]$

$$\begin{aligned}
\oint_{\partial S} \vec{F} \cdot d\vec{r} &= \int_0^{2\pi} \vec{F} \cdot \vec{r}'(t)\ dt \\
&= \int_0^{2\pi} [\sin^3(t), -\cos^3(t), 0] \cdot [-\sin(t), \cos(t), 0]\ dt \\
&= \int_0^{2\pi} (-\sin^4(t) - \cos^4(t))\ dt \\
&= -\int_0^{2\pi} \frac{3}{4} + \frac{1}{4}\cos(4t)\ dt \quad \text{from hint} \\
&= -\left[\frac{3}{4}t + \frac{1}{16}\sin(4t)\right]_0^{2\pi} \\
&= -\frac{3}{4}(2\pi) = -\frac{3}{2}\pi.
\end{aligned}$$

Thus, Stokes' Theorem has been verified. □

Problem 5. ***Conceptual Questions.***

1. *While modeling 1D standing waves on a perfectly elastic string initialy stretched to some length L, what is the expression for c in terms of the tension T and μ — the uniform mass per unit length?*

   Solution. $c = \sqrt{\frac{T}{\mu}}$. □

2. *For the Bessel function of the first kind of zeroth order $J_0(s)$, what is the value of $J_0(\alpha_5)$?*

   Solution. $J_0(\alpha_5) = 0$. □

3. *The solution to the heat equation for a very long bar is written as*

$$u(x,t) = \int_0^{\infty} [A(p)\cos(px) + B(p)\sin(px)]\, \mathrm{e}^{-c^2p^2t}\ dp.$$

   For the initial condition $u(x,0) = f(x)$, write down an expression for $A(p)$.

   Solution. $A(p) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\cos(px)\ dx$ *(similarly,* $B(p) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\sin(px)\ dx$*).* □

Problem 6. *Solve the following PDE using Laplace Transformation:*

$$\frac{\partial w}{\partial x} + x\frac{\partial w}{\partial t} = x,$$

with $w(x,0) = 1$ and $w(0,t) = 1$.

You may use the fact, without proof, that the solution to $\dfrac{\partial W}{\partial x} + xsW = x + \frac{x}{s}$ *is*

$$W = C(s)\mathrm{e}^{-sx^2/2} + \frac{1}{s^2} + \frac{1}{s}.$$

Solution. Using Laplace Transforms on both sides of the PDE:

$$\begin{aligned}
&\mathcal{L}\left(\frac{\partial w}{\partial x} + x\frac{\partial w}{\partial t}\right) = \mathcal{L}(x) \\
\Longrightarrow\ &\mathcal{L}\left(\frac{\partial w}{\partial x}\right) + x\mathcal{L}\left(\frac{\partial w}{\partial t}\right) = x\mathcal{L}(1) \quad \text{Here } x \text{ is treated as a constant} \\
\Longrightarrow\ &W_x + x(sW - w(x,0)) = x\left(\frac{1}{s}\right) \\
\Longrightarrow\ &\frac{\partial W}{\partial s} + sxW - x = \frac{x}{s} \\
\Longrightarrow\ &\frac{\partial W}{\partial s} + sxW = x + \frac{x}{s}.
\end{aligned}$$

Using the given fact, we get

$$W = C(s)\mathrm{e}^{-sx^2/2} + \frac{1}{s^2} + \frac{1}{s}.$$

As $w(0,t) = 1$, we get $W(0,s) = \mathcal{L}(w(0,t)) = \mathcal{L}(1) = \frac{1}{s}$. Then

$$\begin{aligned}
W(0,t) &= C(s)\mathrm{e}^{-s(0)^2/2} + \frac{1}{s^2} + \frac{1}{s} \\
\Longrightarrow\ \frac{1}{s} &= C(s) + \frac{1}{s^2} + \frac{1}{s} \\
\Longrightarrow\ C(s) &= -\frac{1}{s^2}.
\end{aligned}$$

Therefore

$$\begin{aligned}
&W(s,t) = -\frac{1}{s^2}\mathrm{e}^{-sx^2/2} + \frac{1}{s^2} + \frac{1}{s} \\
\Longrightarrow\ &w(x,t) = \mathcal{L}^{-1}(W) = -\mathcal{L}^{-1}\left(\frac{1}{s^2}\mathrm{e}^{-sx^2/2}\right) + \mathcal{L}^{-1}\left(\frac{1}{s^2}\right) + \mathcal{L}^{-1}\left(\frac{1}{s}\right) \\
\Longrightarrow\ &w(x,t) = -\left(t - \frac{1}{2}x^2\right)u\left(t - \frac{1}{2}x^2\right) + t + 1 \\
\Longrightarrow\ &w(x,t) = \begin{cases} t+1 & \text{if } t < \frac{1}{2}x^2 \\ \frac{1}{2}x^2 + 1 & \text{if } t \geq \frac{1}{2}x^2. \end{cases}
\end{aligned}$$

□

Problem 7. *Use separation of variables to find a solution to the PDE*

$$\frac{\partial w}{\partial x} + 2x\frac{\partial w}{\partial t} = 0.$$

No initial or boundary conditions are given.

Solution. Let $w(x,t) = F(x)G(t)$, then from the PDE we get

$$G(t)F'(x) + 2xF(x)\dot{G}(t) = 0.$$

Dividing by $2xF(x)G(t)$, this gives us:

$$\frac{1}{2xF(x)}F'(x) + \frac{1}{G(t)}\dot{G}(t) = 0.$$

Now we separate the variables as $\frac{1}{2xF(x)}F'(x) = k$ and $\frac{1}{G(t)}\dot{G}(t) = -k$. From the first ODE:

$$\begin{aligned}
&\frac{1}{F(x)}\frac{dF}{dx} = 2kx \\
\implies &\ln(F(x)) = kx^2 + C \\
\implies &F(x) = A\mathrm{e}^{kx^2}.
\end{aligned}$$

Then from the second ODE:

$$\begin{aligned}
&\frac{1}{G(t)}\dot{G}(t) = -k \\
\implies &G(t) = B\mathrm{e}^{-kt}.
\end{aligned}$$

Combining the constants into one, we get a solution to the PDE as

$$w(x,t) = F(x)G(t) = Q\mathrm{e}^{kx^2} \cdot \mathrm{e}^{-kt} = Q\mathrm{e}^{k(x^2-t)}.$$

Note: we would need boundary conditions to find Q and initail conditions to find possible restrictions on the values of k. $\square$

Problem 8. *Solve the 2D wave equation $u_{tt} = c^2\nabla^2 u$ on a circular membrane of radius R, where the deformation u is radially symmetric, i.e., $u = u(r,t)$, and satisfies the initial condition $u(r,0) = 2.11J_0\left(\frac{\alpha_5}{R}r\right)$. What are the radii of the (inner) nodal circles for the harmonics? What are the corresponding harmonic frequencies?*

Solution. In this case $f(r) = 2.11J_0\left(\frac{\alpha_5}{R}r\right)$, so $A_5 = 2.11$, and all other $A_m = 0$. Thus, the only mode of vibration is $m = 5$. In that case, there are four inner circles, whose radii are

$$r_1 = \left(\frac{\alpha_1}{\alpha_5}\right)R, \quad r_2 = \left(\frac{\alpha_2}{\alpha_5}\right)R, \quad r_3 = \left(\frac{\alpha_3}{\alpha_5}\right)R, \quad r_4 = \left(\frac{\alpha_4}{\alpha_5}\right)R.$$

There is only one mode, so the harmonic frequency is $f = \frac{\lambda}{2\pi} = \frac{c\alpha_5}{2\pi R}$. $\square$